Research Analysis

Job Sequencing and Tool Switching Problem

Heuristic Bounds, Grouping Selection, and Collapse Variants

Part I Worst-Case Gap: Configuration-Based TSP Heuristic vs. OPT-SSP
Part II Grouping Selection: Mathematical and ML Directions
Part III SSP Variants that Collapse to JGP

Preface on novelty. Several foundational results used here are from the literature and are clearly attributed: the KTNS heuristic and the SSP formulation [1]; NP-hardness of SSP [2]; the LP lower bound $\mathrm{OPT}_{\mathrm{SSP}} \geq m - C$ and column generation for SSP [6]; the Hamiltonian-cycle reformulation of SSP [7]. The *original contributions* of this document are: (i) a tight worst-case bound on the padding-choice gap of the canonical-configuration TSP heuristic with a complete instance construction and proof (section 2); (ii) the equivalence between optimal group ordering and a Max-Weight Hamiltonian Path on the tool-overlap graph (theorem 3.3); and (iii) the general characterisation of cost functions under which SSP collapses to JGP, together with novel structured variants (section 4).

Contents

Notation and Background

Throughout: $J = \{1, \ldots, n\}$ jobs; $T = \{1, \ldots, m\}$ tools; C magazine capacity; $T_j \subseteq T$, $|T_j| \le C$: tool requirement of job j.

A **configuration** for group G_k is any $\mathcal{C}_k \subseteq T$ with $|\mathcal{C}_k| = C$ and $\bigcup_{j \in G_k} T_j \subseteq \mathcal{C}_k$. $\mathfrak{C}(G_k)$: all valid configurations for G_k. Switching cost $w(\mathcal{C}, \mathcal{C}') = |\mathcal{C} \setminus \mathcal{C}'|$ (tools removed = tools inserted under the uniform model [1]).

A **grouping** $\mathcal{G} = \{G_1, \ldots, G_K\}$ is a partition of J into K groups. K^*: JGP-optimal (minimum) number of groups. $s_k = C - |T_{G_k}|$: padding slack of group G_k.

$\mathrm{OPT}_{\mathrm{SSP}}$: optimal SSP cost, minimised over *all* groupings $\mathcal{G}$, all valid configurations $\mathcal{C}_k \in \mathfrak{C}(G_k)$, and all orderings of groups and jobs within groups.

The heuristic under study. (i) Solve JGP to obtain K^* groups. (ii) Assign each group G_k a *canonical* configuration $\hat{\mathcal{C}}_k \in \mathfrak{C}(G_k)$: padding slots are filled without reference to adjacent groups' mandatory tools (e.g. all padding filled by a fixed shared filler set, or independently at random from remaining tools). (iii) Solve TSP on the K^* configurations with arc weight $w(\cdot, \cdot)$ to find the best ordering. (iv) Read off the SSP path. Cost: H.

Where the gap comes from. The TSP in step (iii) finds the optimal *ordering* of the given configurations; it does not alter the configurations themselves. The gap $H - \mathrm{OPT}_{\mathrm{SSP}} \ge 0$ therefore has two sources:

(1) *Padding-choice gap*: the canonical configuration for G_k may waste padding slots on tools that are irrelevant to adjacent groups, forcing more switches than necessary. Choosing padding from adjacent groups' mandatory tools ("lookahead") reduces switches.

(2) *Grouping-choice gap*: OPT-SSP may use a grouping with $K > K^*$ groups, where the extra configuration boundaries allow tool layouts that reduce total switches below anything achievable with K^* groups.

Part I isolates and bounds Source (1) for a fixed JGP-optimal grouping, since that is the gap introduced specifically by the heuristic's padding choice. Source (2) is a property of the JGP/SSP relationship itself and is not attributable to the heuristic; it is discussed separately in remark 2.5.

Part I: Tight Bound on the Padding-Choice Gap

Setup and Key Lemma

For a fixed JGP-optimal grouping $\mathcal{G}^* = \{G_1, \ldots, G_{K^*}\}$ and a fixed ordering π of the groups, the switching cost at each boundary depends on how much of $T_{G_{\pi(k)}}$ survives into $\mathcal{C}_{\pi(k+1)}$. The mandatory tools of $G_{\pi(k)}$ that are *not* required by $G_{\pi(k+1)}$ will survive only if they are placed in the $s_{\pi(k+1)} = C - |T_{G_{\pi(k+1)}}|$ padding slots of $\mathcal{C}_{\pi(k+1)}$.

Lemma 2.1 (Achievable Per-Transition Cost (new)). *For consecutive groups G_k, G_{k+1} in a fixed ordering, the minimum switching cost achievable by any choice of $\mathcal{C}_k \in \mathfrak{C}(G_k)$ and $\mathcal{C}_{k+1} \in \mathfrak{C}(G_{k+1})$ is*

$$w_k^* \;=\; \max\bigl(0,\; |T_{G_k} \setminus T_{G_{k+1}}| - s_{k+1}\bigr) \;=\; \max\bigl(0,\; |T_{G_k} \cup T_{G_{k+1}}| - C\bigr),$$

and it is achieved by the lookahead *configuration $\mathcal{C}_k^* = T_{G_k} \cup X_k$ where $X_k \subseteq T_{G_{k+1}} \setminus T_{G_k}$, $|X_k| = \min(s_k,\, |T_{G_{k+1}} \setminus T_{G_k}|)$.*

Proof. $\mathcal{C}_{k+1}$ must allocate $|T_{G_{k+1}}|$ slots to mandatory tools, leaving s_{k+1} padding slots. Of the $|T_{G_k} \setminus T_{G_{k+1}}|$ tools in $\mathcal{C}_k$ that G_{k+1} does not need, at most s_{k+1} can be kept in $\mathcal{C}_{k+1}$ as padding. The remaining $\max(0, |T_{G_k} \setminus T_{G_{k+1}}| - s_{k+1})$ must leave, giving the lower bound. Equality: set X_k to be the first $\min(s_k, r - ov)$ tools of $T_{G_{k+1}} \setminus T_{G_k}$ where $r = |T_{G_k}|$; these survive into $\mathcal{C}^*_{k+1}$ since $X_k \subseteq T_{G_{k+1}}$. Direct computation gives $|\mathcal{C}^*_k \setminus \mathcal{C}^*_{k+1}| = w^*_k$. □ □

Remark 2.2. The global lower bound $\mathrm{OPT}_{\mathrm{SSP}} \geq m - C$ due to [6] (every tool beyond the initial magazine load must be inserted at least once) is a bound on the SSP over all groupings. lemma 2.1 is a *per-transition* bound for a fixed grouping and ordering; it is a different and complementary tool.

The Tight Construction

Construction 2.3 (Instance $\mathcal{I}(K, r, ov, C)$)**. Parameters.** Positive integers satisfying

$$0 \leq ov \leq \tfrac{r}{2}, \quad r < C, \quad 2r - ov > C. \tag{1}$$

Consistency: $ov \leq r/2$ and $2r - ov > C > r$ require $r > C/2$; e.g. $r = 6$, $ov = 2$, $C = 9$ satisfies $2(6) - 2 = 10 > 9 > 6$ and $2 \leq 3$.

Tool universe. $T = \{t_1, \ldots, t_m\}$, $m = (K-1)(r - ov) + r$.

Mandatory tool sets. For $k = 1, \ldots, K$:

$$T_{G_k} = \{t_{(k-1)(r-ov)+1}, \ldots, t_{(k-1)(r-ov)+r}\},$$

consecutive windows of width r shifted by $r - ov$. Thus $|T_{G_k}| = r$, $|T_{G_k} \cap T_{G_{k+1}}| = ov$, $|T_{G_k} \cup T_{G_{k+1}}| = 2r - ov > C$.

Jobs. G_k contains one job per tool subset of T_{G_k} (or simply one job requiring all of T_{G_k}); the mandatory tool set of the group is T_{G_k}. Padding slack $s = C - r > 0$ for all groups.

JGP-optimality ($K^* = K$)**.** Since $|T_{G_k} \cup T_{G_{k+1}}| > C$ for all k, no configuration covers two adjacent groups, so $K^* = K$.

Heuristic Cost

Use canonical configuration $\hat{\mathcal{C}}_k = T_{G_k} \cup F$ where $F \subseteq T \setminus \bigcup_k T_{G_k}$ is a fixed filler set of s tools disjoint from all mandatory sets. (If $m = Kr - (K-1)ov$ leaves no room for such F, use $F_k \subseteq T \setminus (T_{G_{k-1}} \cup T_{G_k} \cup T_{G_{k+1}})$; for large enough K this is non-empty.) Then for consecutive groups in the ordering:

$$\hat{\mathcal{C}}_k \setminus \hat{\mathcal{C}}_{k+1} = \underbrace{(T_{G_k} \setminus T_{G_{k+1}})}_{r-ov\ \text{tools}} \cup \underbrace{(F \setminus \hat{\mathcal{C}}_{k+1})}_{F \cap T_{G_{k+1}} = \emptyset,\ \text{so all } s \text{ tools leave}},$$

giving $w(\hat{\mathcal{C}}_k, \hat{\mathcal{C}}_{k+1}) = (r - ov) + s = C - ov$ per transition. The TSP orders the K configurations optimally; since all pairwise costs are symmetric and equal by the window structure, any Hamiltonian path has the same cost:

$$H = (K-1)(C - ov). \tag{2}$$

Optimal Cost for This Grouping Under Lookahead Configurations

For the *same* JGP-optimal K-group partition, use lookahead configuration $\mathcal{C}^*_k = T_{G_k} \cup X_k$ with $X_k \subseteq T_{G_{k+1}} \setminus T_{G_k}$, $|X_k| = s = C - r$. This is valid since $|T_{G_{k+1}} \setminus T_{G_k}| = r - ov > s$ (by $2r - ov > C$, i.e. $r - ov > C - r = s$).

Transition cost in the natural ordering $G_1 \to G_2 \to \cdots \to G_K$:

$$\mathcal{C}_k^* \setminus \mathcal{C}_{k+1}^* = (T_{G_k} \setminus \mathcal{C}_{k+1}^*) \cup (\underbrace{X_k \setminus \mathcal{C}_{k+1}^*}_{=\emptyset,\ X_k \subseteq T_{G_{k+1}}}).$$

Now $\mathcal{C}_{k+1}^* = T_{G_{k+1}} \cup X_{k+1}$ with $X_{k+1} \subseteq T_{G_{k+2}} \setminus T_{G_{k+1}}$. Since $ov \le r/2$, windows at distance 2 are disjoint: $T_{G_k} \cap T_{G_{k+2}} = \emptyset$. Hence $T_{G_k} \cap \mathcal{C}_{k+1}^* = T_{G_k} \cap T_{G_{k+1}} = ov$ tools, so

$$|\mathcal{C}_k^* \setminus \mathcal{C}_{k+1}^*| = r - ov.$$

Total cost of the lookahead solution on this grouping:

$$\mathrm{Cost}_{\mathrm{lookahead}} = (K-1)(r-ov). \tag{3}$$

This is an *upper bound* on $\mathrm{OPT}_{\mathrm{SSP}}$:

$$\mathrm{OPT}_{\mathrm{SSP}} \le (K-1)(r-ov). \tag{4}$$

Lower Bound on $\mathrm{OPT}_{\mathrm{SSP}}$ for This Instance

We now prove the matching lower bound.

Lemma 2.4 (Lower Bound via Mandatory Tool Exits)**.** *For instance $\mathcal{I}(K, r, ov, C)$, any feasible SSP solution—over all possible groupings, all valid configurations, and all job orderings—incurs at least $(K-1)(r-ov)$ tool switches.*

Proof. **Define the mandatory exit sets.** For $k = 1, \ldots, K-1$ let

$$D_k = T_{G_k} \setminus T_{G_{k+1}} = \{t_{(k-1)(r-ov)+1}, \ldots, t_{(k-1)(r-ov)+(r-ov)}\},$$

the $r - ov$ tools required by jobs in G_k but by no job in $G_{k+1}, G_{k+2}, \ldots, G_K$. This last claim follows from the window structure: tools in D_k lie at positions $(k-1)(r-ov)+1$ through $k(r-ov)$ in the global tool list, strictly to the left of all windows $G_{k+1}, \ldots, G_K$. The sets $D_1, \ldots, D_{K-1}$ are pairwise disjoint by construction.

Each tool in D_k causes at least one switch in any feasible solution. Fix any tool $t \in D_k$ and any feasible SSP solution (any grouping, any job sequence).

- t is required by at least one job $j \in G_k$. Hence t must be present in the magazine when j is processed. Since the magazine starts with some initial load of C tools, t is either in the initial load (zero insertions so far) or must be inserted at some point before j is processed (one insertion).
- After the last job requiring t is processed (and $t \in D_k$ means no job outside G_k requires t), tool t is never needed again. Because the magazine has finite capacity C and the SSP continues processing further jobs that may require tools not currently loaded, t will be displaced from the magazine at some point. That displacement is one *removal*.
- In the uniform SSP model [1], the magazine stays at capacity C throughout: every removal is paired with an insertion and vice versa. So either t was in the initial load and gets removed once (one switch), or t was not in the initial load, gets inserted once and removed once (two switches). In both cases, t contributes at least **one switch** (the removal) to the total switch count.

This argument is independent of the grouping used. No matter how the solution groups and sequences jobs, the requirement that $t \in D_k$ must be present when some job in G_k runs, and absent (eventually) afterwards, is a property of the *instance* (which jobs require which tools), not of the solution structure. A finer grouping (more configurations) does not change which tools are required by which jobs.

Count. Since $D_1, \ldots, D_{K-1}$ are pairwise disjoint, the $\sum_{k=1}^{K-1} |D_k| = (K-1)(r-ov)$ removal events for tools in $\bigcup_k D_k$ are all distinct switch events. Hence total switches $\geq (K-1)(r-ov)$. □ □

Remark 2.5 (On Source (2) of the gap)**.** lemma 2.4 shows that $\mathrm{OPT}_{\mathrm{SSP}} = (K-1)(r-ov)$ for this instance: the lower bound matches the lookahead upper bound (4). Importantly, the lower bound holds for any grouping, including those with $K' > K$ groups: a finer grouping cannot reduce the mandatory removal count of tools in $\bigcup_k D_k$, because tool requirements are fixed by the instance. Hence for $\mathcal{I}(K, r, ov, C)$, Source (2) of the gap (grouping choice) contributes *zero*: the optimal SSP uses exactly $K = K^*$ groups, and the entire gap is Source (1) (padding choice). This makes the construction a clean isolator of Source (1).

Main Result of Part I

Theorem 2.6 (Tight Padding-Choice Gap)**.** *For instance $\mathcal{I}(K, r, ov, C)$ with parameters satisfying (1):*

$$\begin{aligned} H &= (K-1)(C-ov), \\ \mathrm{OPT}_{\mathrm{SSP}} &= (K-1)(r-ov), \\ H - \mathrm{OPT}_{\mathrm{SSP}} &= (K-1)(C-r) = (K-1)\, s, \end{aligned}$$

*where $s = C - r$ is the per-group padding slack. The bound is **tight**: it is achieved by the construction above. No canonical-padding TSP heuristic can guarantee a smaller gap on this family.*

Proof. H follows from (2). $\mathrm{OPT}_{\mathrm{SSP}} = (K-1)(r-ov)$ follows from the upper bound (4) and lemma 2.4. The gap $= (K-1)(C-ov) - (K-1)(r-ov) = (K-1)(C-r)$. Tightness: the heuristic cost (2) is a worst-case cost for canonical padding (any filler F disjoint from adjacent mandatory sets gives the same cost); no canonical strategy does better since it cannot know adjacency. □ □

Corollary 2.7 (Unbounded Approximation Ratio)**.**

$$\frac{H}{\mathrm{OPT}_{\mathrm{SSP}}} = \frac{C-ov}{r-ov} = 1 + \frac{s}{r-ov}.$$

As $s \to r - ov - 1$ (slack approaches maximum under the constraints), the ratio approaches $C - ov$, which grows with C. As $r - ov \to 1$, the ratio approaches $C - ov + 1 \approx C$. The ratio is unbounded as $C \to \infty$: ***no constant approximation factor*** *exists for the canonical-padding TSP heuristic.*

Remark 2.8 (Gap is zero when $s = 0$)**.** When $r = C$ (jobs fill the magazine completely), no padding freedom exists and every configuration is forced. SSP then reduces to a pure TSP on mandatory tool sets, consistent with the known case where $|T_j| = C$ for all j [7].

Part II: Grouping Selection

Context. You have: all JGP-optimal groupings ($K = K^*$), some sub-optimal ones ($K > K^*$), and the trivial all-configurations option. The goal: select groupings to (a) solve SSP exactly or (b) obtain strong heuristic solutions with quality guarantees.

Mathematical Directions

GTSP Formulation (Reference and Extension)

Known. Ghiani et al. [7] showed that SSP is equivalent to a minimum-cost Hamiltonian cycle on a graph of all valid configurations. Column generation for SSP with set-covering master and bin-packing pricing was developed in [6].

Extension to fixed grouping (new framing). For a *fixed* grouping $\mathcal{G} = \{G_1, \ldots, G_K\}$, the subproblem of jointly selecting configurations and ordering groups is a Generalised TSP (GTSP): clusters are $V_k = \mathfrak{C}(G_k)$, arc costs are $w(\mathcal{C}_i, \mathcal{C}_j)$. This is the natural decomposition when iterating over candidate groupings.

Theorem 3.1 (GTSP Equivalence for Fixed Grouping)**.** *For a fixed grouping $\mathcal{G}$, $\mathrm{OPT}(\mathcal{G})$ = minimum-cost Hamiltonian path in the GTSP on clusters $\{V_k\}_{k=1}^K$.*

Proof. Any feasible SSP path on $\mathcal{G}$ selects one configuration per group and an ordering, corresponding to exactly a Hamiltonian path visiting one node per cluster. Minimising over both choices is the GTSP. □ □

Solving GTSP. The Noon–Bean transformation [8] reduces GTSP to standard TSP on $O(N)$ nodes ($N = \sum_k |V_k|$); Concorde or LKH3 apply directly. Held–Karp DP for GTSP runs in $O(N^2 \cdot 2^K)$, tractable for small K^*.
Cluster size. $|V_k| = \binom{m-|T_{G_k}|}{s_k}$ is exponential in s_k. Restrict to *structured configurations*: $\mathcal{C}_k = T_{G_k} \cup X_k$ with $X_k \subseteq T_{G_{k+1}} \setminus T_{G_k}$ (lookahead, section 2.2); this limits each cluster to $O(r - ov)$ nodes and recovers the tight construction of Part I as the GTSP optimum.

Optimal Ordering via Tool-Overlap Graph (New Result)

Definition 3.2 (Tool-Overlap Graph)**.** For grouping $\mathcal{G} = \{G_1, \ldots, G_K\}$, define the complete weighted graph H_{ov} on K nodes with edge weight $w(G_k, G_{k'}) = |T_{G_k} \cap T_{G_{k'}}|$.

Theorem 3.3 (Optimal Ordering = Max-Weight Hamiltonian Path (New))**.** *For a fixed grouping with $|T_{G_k}| = r$ for all k and padding slack $s \geq r - \min_k |T_{G_k} \setminus T_{G_{k+1}}|$ (sufficient to enable full lookahead), the minimum SSP cost over all orderings is:*

$$\mathrm{OPT}(\mathcal{G}, \pi^*) = (K-1)r - \max_{\pi} \sum_{k=1}^{K-1} |T_{G_{\pi(k)}} \cap T_{G_{\pi(k+1)}}|.$$

Hence the optimal ordering π^* solves a Max-Weight Hamiltonian Path (MWHP) on H_{ov}, solvable in $O(K^2 \cdot 2^K)$ by Held–Karp DP.

Proof. Under lookahead configuration $\mathcal{C}^*_{\pi(k)} = T_{G_{\pi(k)}} \cup X_{\pi(k)}$ with $X_{\pi(k)} \subseteq T_{G_{\pi(k+1)}} \setminus T_{G_{\pi(k)}}$, $|X_{\pi(k)}| = s$, all padding tools survive the transition (they belong to $T_{G_{\pi(k+1)}}$), and the mandatory overlap $T_{G_{\pi(k)}} \cap T_{G_{\pi(k+1)}}$ also survives. Thus $|\mathcal{C}^*_{\pi(k)} \setminus \mathcal{C}^*_{\pi(k+1)}| = |T_{G_{\pi(k)}}| - |T_{G_{\pi(k)}} \cap T_{G_{\pi(k+1)}}| = r - w(G_{\pi(k)}, G_{\pi(k+1)})$. Summing over $k = 1, \ldots, K-1$ and using $\sum_k r = (K-1)r = \text{const}$, minimising total cost is equivalent to maximising $\sum_k w(G_{\pi(k)}, G_{\pi(k+1)})$. □ □

Practical use. Enumerate candidate groupings; for each, build H_{ov} and solve MWHP (tractable for small K^*); score groupings by MWHP value; solve GTSP exactly for the top-p groupings; return the best. The MWHP score is a principled, polynomial-time proxy for $\mathrm{OPT}(\mathcal{G})$ under the lookahead assumption.

Lagrangian Relaxation for the JGP/SSP Tradeoff

The JGP/SSP tension is captured by

$$L(\lambda) = \min_{\mathcal{G},\pi,\text{configs}} \left[\text{cost}(\mathcal{G}, \pi) + \lambda(|\mathcal{G}| - K^*)\right], \quad \lambda \geq 0.$$

By weak Lagrangian duality [9, Ch. 10]: $L(\lambda) \leq \text{OPT}_{\text{SSP}}$ for all $\lambda \geq 0$. The subgradient algorithm (standard; see [9]) traces the Pareto frontier of (number of groups, switch cost), providing sub-optimal JGP groupings ($K > K^*$) that may achieve lower switching cost than any K^*-group solution. The tightest bound is $L(\lambda^*) = \max_{\lambda \geq 0} L(\lambda)$; the gap $\text{OPT}_{\text{SSP}} - L(\lambda^*)$ equals the LP integrality gap for the joint problem.

Column Generation (Reference + Grouping-Selection Use)

Column generation for SSP was developed in [6] with a set-covering master over feasible groups and a bin-packing pricing subproblem. We use it here in a complementary way: rather than solving SSP from scratch, we use the LP dual variables u_j from an intermediate CG iteration as a *scoring function* for candidate groupings. A grouping $\mathcal{G}$ with high $\sum_{G_k \in \mathcal{G}} \sum_{j \in G_k} u_j$ is "dual-cheap" and should be prioritised for exact GTSP evaluation. This provides a data-driven ranking of candidate groupings compatible with the CG framework, without requiring a full branch-and-price run.

Tool-Conflict Graph and Chromatic Structure

Definition 3.4 (Tool Conflict Graph (standard; see [2]))**.** $G_c = (J, E_c)$ where $(j, j') \in E_c \iff |T_j \cup T_{j'}| > C$.

Valid groupings correspond to proper K^*-colourings of G_c (each colour class is a feasible group). Among all K^*-colourings, the MWHP criterion of theorem 3.3 selects the one whose colour classes have maximum pairwise mandatory tool overlap in the optimal ordering. This connects the chromatic structure of G_c to the SSP objective in a way not, to our knowledge, made explicit in prior work.

Machine Learning Directions

GNN Grouping Scorer

Goal. Score a large candidate pool without solving GTSP for each grouping.

Architecture. Encode the instance as a bipartite job–tool graph $B = (J \cup T, E)$ with $(j, t) \in E \iff t \in T_j$. A two-layer Graph Attention Network (GAT) or GraphSAGE produces job embeddings $h_j \in \mathbb{R}^d$. For grouping $\mathcal{G}$, represent group G_k as $\mathbf{g}_k = \frac{1}{|G_k|} \sum_{j \in G_k} h_j$. A second GNN on the K-node grouping graph (edge features $|T_{G_k} \cap T_{G_{k'}}|$) predicts $\text{OPT}(\mathcal{G})$.

Justification. $\text{OPT}(\mathcal{G})$ depends on the tool-overlap structure of the grouping, captured by the K-node graph. GNNs are permutation-equivariant (the SSP cost is permutation-invariant in groups) and are universal approximators on graphs [10]; hence the architecture is principled.

Training. Solve instances up to moderate scale exactly; include both $K = K^*$ and $K > K^*$ groupings so the model learns across the Pareto frontier. Loss: $\mathbb{E}[|\hat{c}(\mathcal{G}) - \text{OPT}(\mathcal{G})|^2]$.

Inference. Score all candidates in $O(K^2)$ per grouping; solve GTSP for top-p; return best.

Guarantee. If the GNN approximates $\text{OPT}(\mathcal{G})$ within error ε w.p. $\geq 1 - \delta$, then the returned solution has cost $\leq \text{OPT}_{\text{SSP}} + \varepsilon + \eta$, where $\eta \to 0$ as $p \to \infty$.

Reinforcement Learning for Joint Grouping and Sequencing

RL naturally explores $K > K^*$, learning when more groups reduce total switches without being constrained to any fixed grouping size.

MDP. State: partial job-to-group assignment. Action: assign next job to an existing feasible group or open a new one (masked to maintain $|T_{G_k}| \le C$). Terminal reward $R_T = -\text{cost}(\text{final grouping, GTSP-optimal configs})$.

Training. REINFORCE with greedy rollout baseline [12]:

$$\nabla_\theta \mathcal{L} = -\mathbb{E}[(R_T - b)\nabla_\theta \log \pi_\theta].$$

Architecture: Transformer over job feature vectors $[|T_j|, \phi_j]$ ($\phi_j \in \{0,1\}^m$ the tool-requirement vector) following [12].

Distinction from JGP solvers. A pure JGP solver minimises K; the RL policy minimises switching cost, discovering that $K > K^*$ can be beneficial. The reward directly measures switching cost, not group count.

Guarantee. Distributional: $\mathbb{E}_{I\sim\mathcal{D}}[\text{cost}(\pi_\theta(I))] \to \mathbb{E}_{I\sim\mathcal{D}}[\text{OPT}_{\text{SSP}}(I)]$ under standard RL convergence conditions. No worst-case bound.

Learning-Augmented Column Generation

Setup. CG for SSP [6] is exact but the pricing problem is costly. Train a predictor $P_\phi : (u, B) \to G'$ (dual variables u and instance graph B to a candidate feasible group G') on (u^*, G^*) pairs from solved instances.

Protocol. At each CG iteration: query P_ϕ; verify reduced cost $c_{G'} - \sum_{j\in G'} u_j$ (cheap); if negative, add to restricted master; else fall back to exact pricing.

Correctness. Every ML-suggested column is verified before use; exact LP optimality is certified by the fallback solver. Exact optimality is preserved; ML only reduces the number of exact pricing calls. Analogous approaches reduce pricing iterations by 50–80% on related problems [13].

Contrastive Siamese GNN for Grouping Ranking

Train a Siamese GNN on pairs $(\mathcal{G}_A, \mathcal{G}_B)$ labelled by which has lower OPT. Loss: binary cross-entropy. At inference: tournament over N candidates in $O(N \log N)$ comparisons. Only the top-ranked grouping is passed to GTSP. Requires only *relative* quality labels, cheaper to obtain than exact costs.

Part III: SSP Variants That Collapse to JGP

Definition 4.1 (Collapse to JGP)**.** SSP-f (SSP with cost function $f : 2^T \times 2^T \to \mathbb{R}_+$) *collapses to JGP* if for every instance I, $\text{OPT}_{\text{SSP-}f}$ is achieved by some grouping with exactly $K^*(I)$ groups.

Baseline: Switching Instants (Known)

$f_{01}(\mathcal{C}, \mathcal{C}') = \mathbf{1}[\mathcal{C} \neq \mathcal{C}']$: cost equals number of configuration changes $= K - 1$ for K groups. Minimising $\Rightarrow$ JGP. This is the Tool Switching Instants Problem (TSIP), known to be equivalent to JGP [1]. It serves as the baseline; we seek metrically richer variants.

General Collapse Characterisation (New)

Rather than analysing specific variants individually, we first characterise all functions f that cause collapse. This is new and subsumes the threshold and weighted variants as corollaries.

Theorem 4.2 (Necessary and Sufficient Conditions for Collapse). *f causes SSP-f to collapse to JGP on all instances if and only if:*

(C1) Self-cost zero: $f(\mathcal{C}, \mathcal{C}) = 0$ *for all* $\mathcal{C}$.

(C2) Strict inter-group positivity: *For any JGP-optimal grouping, any two groups* $G_k \neq G_{k'}$ *in it, and any valid configurations* $\mathcal{C} \in \mathfrak{C}(G_k)$, $\mathcal{C}' \in \mathfrak{C}(G_{k'})$: $f(\mathcal{C}, \mathcal{C}') \geq \delta_{\min} > 0$.

(C3) No-beneficial-split: *Adding any extra group boundary (splitting a group into two) does not reduce the SSP-f cost. Formally: for every feasible grouping* $\mathcal{G}$ *with* $K > K^*$ *groups,* $\mathrm{OPT}_f(\mathcal{G}) \geq \mathrm{OPT}_f(\mathcal{G}')$ *for some JGP-optimal* $\mathcal{G}'$ *with* K^* *groups.*

Proof. **Necessity.**
$\neg$(C1): If $f(\mathcal{C}, \mathcal{C}) > 0$ for some $\mathcal{C}$, then repeated uses of $\mathcal{C}$ incur positive cost, which is unphysical (no switch occurs). More concretely: within a single group's processing, the magazine does not change, so any well-defined switching cost must assign zero cost to "no change". If (C1) fails, f models a cost even for non-events, and the SSP-f objective no longer corresponds to any meaningful switching count. Any valid switching cost function must satisfy (C1).

$\neg$(C2): Suppose $f(\mathcal{C}, \mathcal{C}') = 0$ for some $\mathcal{C} \in \mathfrak{C}(G_k)$, $\mathcal{C}' \in \mathfrak{C}(G_{k'})$, $G_k \neq G_{k'}$. Split G_k into two sub-groups $G_k^{(1)}, G_k^{(2)}$ with the same configuration $\mathcal{C}$ for both, and insert $\mathcal{C}'$ between them. The transition $\mathcal{C} \to \mathcal{C}' \to \mathcal{C}$ costs $0 + 0 = 0$. This produces a $(K^* + 1)$-group solution with the same SSP-f cost as the K^*-group solution, so the $(K^* + 1)$-group solution is also optimal — collapse fails.

$\neg$(C3): By definition, if some $K > K^*$ grouping achieves strictly lower f-cost than all K^*-group solutions, the SSP-f optimum uses $K > K^*$ groups, violating definition 4.1.

Sufficiency. Under (C2), any solution with K groups incurs at least $K - 1$ inter-group transitions, each costing $\geq \delta_{\min} > 0$, giving total cost $\geq (K - 1)\delta_{\min}$. For $K > K^*$: $(K - 1)\delta_{\min} > (K^* - 1)\delta_{\min}$. The K^*-group optimum $c^*_{K^*}$ satisfies $c^*_{K^*} \geq (K^* - 1)\delta_{\min}$. Under (C3), $c^*_{K^*} \leq \mathrm{OPT}_f(K)$ for all $K > K^*$. Hence every optimal solution uses exactly K^* groups. ☐ ☐

Remark 4.3. Condition (C3) is the hardest to verify in general. The threshold and setup-matrix variants below provide concrete sufficient conditions for (C3) in terms of instance parameters.

Threshold SSP (New Variant)

Definition 4.4 (Threshold SSP). For $\theta \geq 0$: $f_\theta(\mathcal{C}, \mathcal{C}') = \max(0, |\mathcal{C} \setminus \mathcal{C}'| - \theta)$. Switching $\leq \theta$ tools is free; excess switches incur unit cost.

Theorem 4.5 (Threshold Collapse). *For a JGP-optimal grouping* $\mathcal{G}^*$ *with lookahead configurations, define*

$$\theta^*(\mathcal{G}^*) = \max_{k=1,\ldots,K^*-1} |T_{G_k} \setminus T_{G_{k+1}}|.$$

For $\theta \geq \theta^*(\mathcal{G}^*)$: *SSP-$f_\theta$ achieves cost* 0 *on* $\mathcal{G}^*$ *and collapses to JGP. For* $0 \leq \theta < \theta^*(\mathcal{G}^*)$:

SSP-f_θ has strictly positive optimal cost and interpolates between SSP and TSIP.

Proof. Collapse: Each lookahead transition costs $|T_{G_k} \setminus T_{G_{k+1}}| \le \theta^* \le \theta$, so $f_\theta(\mathcal{C}_k^*, \mathcal{C}_{k+1}^*) = 0$ for all k. Total cost $= 0 = \mathrm{OPT}_{\text{SSP-}f_\theta}$ (since $f_\theta \ge 0$). Condition (C2) of theorem 4.2 holds with $\delta_{\min} = 1$ (any transition between groups using different configurations costs at least $1 - \theta \cdot 0 = 1$ after thresholding if $\theta = 0$; more generally $\delta_{\min} = \max(0, \min_{\text{inter-group}} w(\cdot,\cdot) - \theta)$ which is positive when $\theta < \min_{\text{inter-group}} w(\cdot,\cdot)$, i.e. below the minimum inter-group switching cost, which is > 0 in any JGP-optimal grouping). Condition (C3) holds because the 0-cost K^*-group solution cannot be improved.

Non-collapse for $\theta < \theta^*$: There exists a consecutive pair (G_{k^*}, G_{k^*+1}) with $|T_{G_{k^*}} \setminus T_{G_{k^*+1}}| > \theta$. By lemma 2.1, the minimum transition cost for this pair over all valid configurations is $w_{k^*}^* = |T_{G_{k^*}} \cup T_{G_{k^*+1}}| - C > 0$. After thresholding: $f_\theta \ge \max(0, w_{k^*}^* - \theta) > 0$ for $\theta < w_{k^*}^*$. Hence $\mathrm{OPT}_{\text{SSP-}f_\theta} > 0$. □ □

Remark 4.6. The collapse threshold $\theta^*(\mathcal{G}^*)$ depends on the grouping. The grouping with highest inter-group tool overlap (selected by theorem 3.3) minimises θ^* and makes collapse easier to achieve in practice.

Setup-Matrix SSP (New Variant)

Definition 4.7 (Setup-Matrix SSP)**.** Let $W \in \mathbb{R}_+^{m \times m}$ with $W_{tt} = 0$ and $W_{tt'} \ge 0$ for $t \ne t'$. $f_W(\mathcal{C}, \mathcal{C}') = \sum_{t \in \mathcal{C} \setminus \mathcal{C}'} \min_{t' \in \mathcal{C}' \setminus \mathcal{C}} W_{tt'}$. Here $W_{tt'}$ is the cost of removing t when t' is loaded in its place, modelling sequence-dependent slot-reuse costs.

Theorem 4.8 (Setup-Matrix Collapse Condition)**.** *If $W_{tt'} \ge \gamma > 0$ for all $t \ne t'$, then for any consecutive group pair and any valid configurations: $f_W(\mathcal{C}_k, \mathcal{C}_{k+1}) \ge \gamma \cdot w_k^*$, where $w_k^* = \max(0, |T_{G_k} \cup T_{G_{k+1}}| - C)$ from lemma 2.1. Condition (C2) holds with $\delta_{\min} = \gamma \cdot \min_k w_k^* > 0$. SSP-$f_W$ collapses to JGP when additionally (C3) holds: $\gamma \cdot w_k^*$ exceeds the cost reduction from splitting any group.*

Proof. Each removed tool $t \in \mathcal{C}_k \setminus \mathcal{C}_{k+1}$ incurs cost $\min_{t' \in \mathcal{C}_{k+1} \setminus \mathcal{C}_k} W_{tt'} \ge \gamma$ (the set $\mathcal{C}_{k+1} \setminus \mathcal{C}_k$ is non-empty and all $t' \ne t$). Summing: $f_W \ge \gamma |\mathcal{C}_k \setminus \mathcal{C}_{k+1}| \ge \gamma \cdot w_k^*$. For inter-group pairs, $w_k^* > 0$ (adjacent groups cannot share a configuration in a JGP-optimal grouping), so (C2) holds. □ □

Machine Learning for Collapse Variants

Learning the Collapse Threshold

Train a regressor on instance features $(n, m, C, \mathbb{E}[|T_j|], \mathrm{Var}[|T_j|], \max |T_j|, \rho)$ (where ρ is tool co-occurrence density) to predict θ^*. At inference: predict $\hat{\theta}^*$; solve SSP-$f_{\hat{\theta}^*}$; if accurate, the solution is JGP-optimal with zero threshold-cost.

Meta-Learning a Collapse-Inducing Cost Function (New)

Learn a parameterised $f_\phi = f_{W_\phi}$ (setup-matrix variant) satisfying both collapse and fidelity to the original SSP:

$$\min_\phi \sum_I |\mathrm{OPT}_{\text{SSP-}f_\phi}(I) - \mathrm{OPT}_{\text{SSP}}(I)| + \lambda R(\phi), \tag{5}$$

where $R(\phi) = \sum_I \max(0, \mathrm{OPT}_{f_\phi}(I, K \le K^*) - \mathrm{OPT}_{f_\phi}(I, K = K^*))$ enforces collapse ($R(\phi) = 0 \iff$ SSP-f_ϕ collapses to JGP on training instances).

This is surrogate optimisation: SSP-f_ϕ (tractable via collapse) approximates the original SSP (hard), with the two objectives in (5) trading off tractability against fidelity. Convergence guarantees and sample complexity for this bilevel formulation are open.

Lagrangian Parameter Prediction

Rather than running the full subgradient algorithm of section 3.1.3 on each new instance, train a model $\hat{\lambda}^*(I) = f_\psi(\text{features}(I))$ from solved instances [11]. Predict $\hat{\lambda}^*$; solve $L(\hat{\lambda}^*)$ in one pass. Analogous warm-starting approaches reduce solve times by 50–90% in MIP settings.

Summary

Table 1: Key results: origin and status.

Part	Result	Basis	Origin
I	Gap $H - \text{OPT} = (K-1)s$; tight construction and full proof	Mandatory-exit LB (lemma 2.4) + lookahead UB; both tight	**New**
I	Ratio $H/\text{OPT} = (C - ov)/(r - ov)$; unbounded	Follows from construction	**New**
I	Gap $= 0$ when $s = 0$; SSP reduces to TSP on mandatory sets	Consistent with [7]	Known
II	GTSP for fixed grouping	Specialisation of [7] to fixed $\mathcal{G}$	Framing new
II	Optimal ordering = MWHP on overlap graph (theorem 3.3)	New equivalence; tractable via Held–Karp	**New**
II	Lagrangian for JGP/SSP Pareto frontier	Standard Lagrangian duality [9]; application is new	Application new
II	Column generation for SSP	[6]; reused here as grouping scorer	Reference
II	GNN grouping scorer; RL for joint grouping+sequencing	Architectures from [10, 12]; application new	**New**
II	Learning-augmented CG; contrastive ranking	Framework from [13]; application new	**New**
III	TSIP $\equiv$ JGP (baseline)	[1]	Known
III	General collapse characterisation (C1)+(C2)+(C3) (theorem 4.2)	New necessary and sufficient conditions	**New**
III	Threshold SSP collapse (theorem 4.5)	Corollary of theorem 4.2; new variant	**New**
III	Setup-Matrix SSP and γ-collapse condition (theorem 4.8)	New variant; new collapse proof	**New**
III	Meta-learning collapse-inducing f_ϕ (bilevel, section 4.5.2)	New open problem; novel direction	**New**

Open Research Questions

Q1. (I) Is the MWHP score on H_{ov} a provably tight proxy for $\text{OPT}(\mathcal{G})$ (not just under full lookahead)? Proving approximation ratios for groupings selected by this criterion is open.

Q2. (II) What is the complexity of the CG pricing problem for SSP? Fixed-parameter tractability in K^* would make branch-and-price practical.

Q3. (II) Can GTSP cluster sizes be polynomially bounded for structured tool families (laminar, interval)? This would make theorem 3.1 directly tractable.

Q4. (III) Does a function f exist that (a) makes SSP-f poly-time, (b) collapses to JGP, and (c) preserves OPT_{SSP} within a constant? Likely no in general (implies P=NP), but open for parameterised versions.

Q5. (III/ML) Convergence and sample complexity for the bilevel formulation (5) are open; the landscape of $R(\phi)$ is non-convex and potentially discontinuous.

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